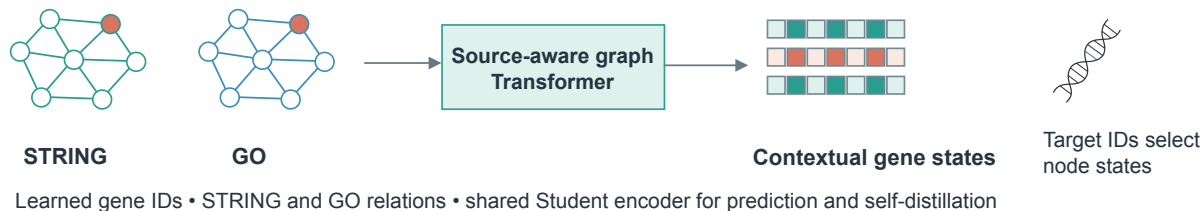


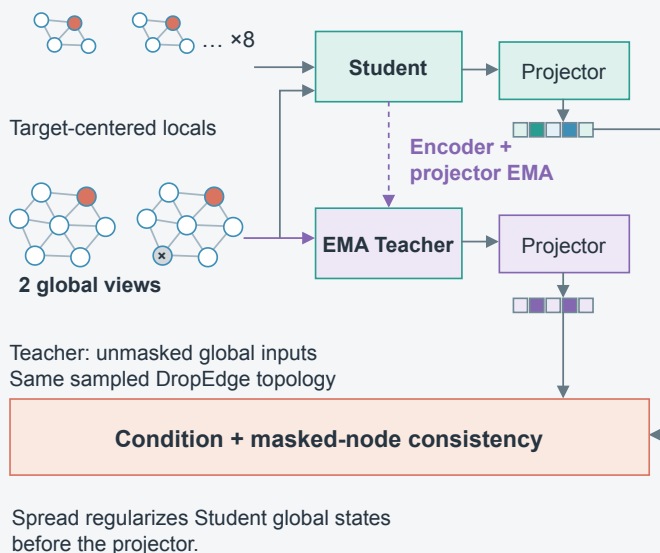
Overview

Joint perturbation prediction and graph self-distillation

a Shared gene representations from public biological context

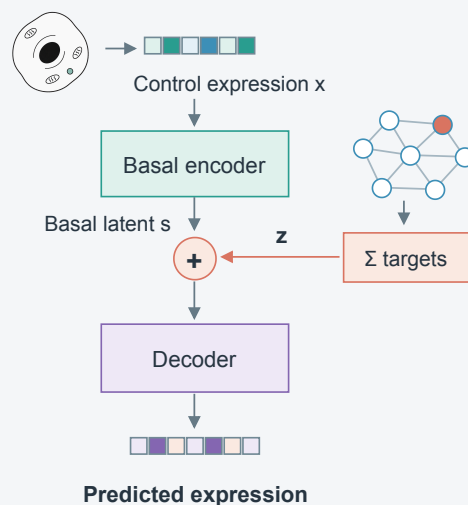


b Learning across graph views

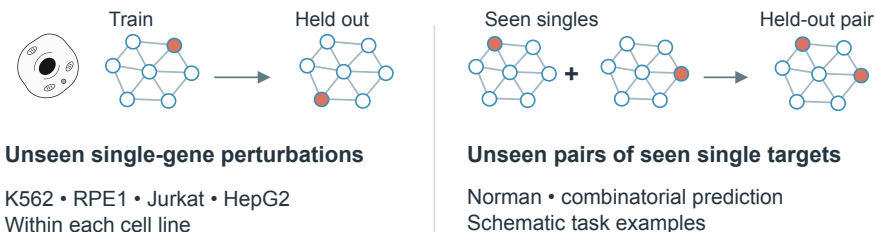


Training: global/local consistency + prediction

c Control-conditioned prediction



d Broader five-dataset benchmark and joint objective



Prediction + self-distillation

$$L = L_{\text{pred}} + 0.8 L_{\text{cond}} + 0.4 L_{\text{mask}} + 0.1 L_{\text{spread}}$$

L_{pred} : MSE against observed perturbed expression